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CFD and ANN surrogate modelling for fixing angle optimisation of a Darrieus vertical axis wind turbine

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ABSTRACT

Global decarbonisation requires efficient and reliable renewable energy systems, with Vertical Axis Wind Turbines (VAWTs) offering advantages in turbulent and urban environments. This study examines the effect of blade Fixing Angle (FA) on the aerodynamic performance of a three-bladed Darrieus VAWT using Unsteady Reynolds-Averaged Navier–Stokes (URANS) based Computational Fluid Dynamics (CFD) simulations coupled with an Artificial Neural Network (ANN) surrogate model. A parametric analysis was carried out for seven FA values across a Tip-Speed Ratio (TSR) range of 0.5 – 3.0. Results show that a small positive FA (toe-out) improves efficiency by delaying stall onset and extending upwind torque production. The optimum case ($FA = +1.5^\circ$, $TSR = 2.0$) achieved a 15.3% increase in power coefficient (C_p) compared to the baseline ($FA = 0^\circ$, $TSR = 2.0$). In contrast, negative FA reduced performance due to earlier separation and greater downwind losses. The ANN surrogate, trained on CFD data, effectively captured the nonlinear dependence of C_p on TSR and FA, achieving $R^2 = 0.98$ and a Mean Absolute Error (MAE) below 0.01. Once validated, the ANN enabled rapid optimisation of the design space, locating the global optimum with negligible computational cost.

Flow-field and force-coefficient analyses further reveal that positive FA modifies the incidence-angle distribution, mitigates torque ripple, and suppresses dynamic stall, thereby explaining the observed performance enhancement. Overall, the integration of CFD and ANN provides both physical insight and computational efficiency, establishing a solid foundation for future multi-objective optimisation of VAWTs considering parameters such as C_p , torque ripple, and self-starting capability.

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Vertical Axis Wind Turbine (VAWT), Darrieus Rotor, Fixing Angle (FA), Tip-Speed Ratio (TSR); Computational Fluid Dynamics (CFD), Artificial Neural Network (ANN), Dynamic Stall, Aerodynamic Optimisation

Nomenclature

A	Rotor swept area (m^2)
BEM	Blade Element Momentum method
C_m	Moment coefficient = $\frac{M}{0.5\rho U^2 AR}$
C	Blade chord length (m)
D	Rotor diameter (m)
DOE	Design of Experiments
FA	Fixing Angle ($^\circ$), preset angular offset of blade mounting (toe-in/out)
H	Blade height (m)
k	Turbulent kinetic energy (m^2/s^2)
M	Rotor moment ($N \cdot m$)
R	Rotor radius (m)
R^2	Coefficient of determination (correlation measure)
SST	Shear Stress Transport turbulence model
TKE	Turbulent Kinetic Energy
U	Freestream velocity (m/s)

ν	Kinematic viscosity (m^2/s)
ρ	Air density (kg/m^3)
ω	Angular velocity (rad/s)
$\Delta\theta$	Azimuthal increment per time-step ($^\circ$)
Δt	Time-step size (s)

1. Introduction

Global warming, primarily caused by increased greenhouse gas emissions from fossil fuel combustion, poses severe environmental and societal challenges. The growing atmospheric concentration of carbon dioxide has accelerated climate change, driving the urgent global shift towards sustainable and renewable energy systems (Cozzi et al., 2020; Masson-Delmotte et al., 2021). Among various renewable energy sources, wind energy has emerged as one of the most promising due to its sustainability, zero emissions, and wide availability (Global Wind

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Energy Council, 2021). Its role in helping countries reduce dependence on fossil fuels and lower their carbon footprints makes it a key pillar of future energy policy.

Wind turbines are generally classified into Horizontal Axis Wind Turbines (*HAWTs*) and Vertical Axis Wind Turbines (*VAWTs*). *HAWTs* dominate utility-scale power generation due to their high efficiency; however, they require complex yaw mechanisms and perform poorly in turbulent and multidirectional winds (Gielen et al., 2019). In contrast, *VAWTs* offer several advantages such as omnidirectional wind capture, simpler mechanical design, easier maintenance, and superior performance in urban or turbulent environments (Bhutta et al., 2012). *VAWTs* are further classified into two main types: drag-based Savonius turbines and lift-based Darrieus turbines. Savonius turbines are aerodynamically inferior because their drag-based energy extraction yields inherently low power coefficients and restricts operation to very low *TSRs*, unlike lift-based Darrieus turbines which achieve higher efficiency (Bravo et al., 2007; Hand & Cashman, 2020; Kingma & Ba, 2014).

Darrieus *VAWTs* exhibit complex aerodynamic characteristics, involving variations in lift and drag forces, changes in Angle of Attack (*AOA*), dynamic stall, and flow separation. Lift acts perpendicular to the direction of relative wind, whereas drag acts parallel to it, opposing the blade's motion (Li et al., 2016). The *AOA*, defined as the angle between the chord line of the blade and the incoming relative wind, varies cyclically due to the blades' cylindrical rotation around the vertical axis, and this variation significantly affects aerodynamic performance (Hand & Cashman, 2020). Dynamic stall occurs due to rapid variations in *AOA*, causing abrupt changes in lift and drag forces, which result in unsteady torque and reduced aerodynamic efficiency (Li et al., 2018). Flow separation, where airflow detaches from the blade surface, exacerbates dynamic stall and further degrades aerodynamic performance (Difuntorum & Danao, 2019). Recent high-fidelity *URANS* simulations, such as (Wang et al., 2024), have further demonstrated the importance of accurately capturing transient flow behaviour in coupled aerodynamic-motion systems, highlighting the necessity of unsteady *CFD* modelling for reliable performance prediction. Aerodynamic behaviour is highly sensitive to blade profile (i.e. aerofoil) (Tangler & Somers, 1995), chord length (Yang, 2020), rotor solidity (Kulak et al., 2024), and rotor diameter (Singh & Santoso, 2011), and *FA* (Difuntorum & Danao, 2019). Among the geometric factors, an especially important parameter is the *FA* – the preset mounting angle of the blade relative to the rotor (Difuntorum & Danao, 2019).

The *FA* alters the azimuthal evolution of the blade *AOA* : a positive *FA* reduces the effective *AOA* during

the upwind half-cycle and increases it in the downwind sector, thereby redistributing aerodynamic loading and modifying lift generation, torque production, and overall power output. Consistent with this mechanism, parametric investigations have shown that a slight toe-out (i.e. positive *FA*, with the leading edge oriented outward from the hub, e.g. $+1^\circ$) enhances performance by delaying stall and extending upwind torque contribution, whereas toe-in (i.e. negative *FA*, with the leading edge oriented inward toward the hub, e.g. -1°) reduces efficiency due to premature separation. For example, Difuntorum & Danao (2019) reported that at *TSR*, $\lambda = 4$, a *FA* = $+1^\circ$ toe-out increased peak C_P by $\sim 0.9\%$, while a *FA* = -1.5° toe-in reduced peak C_P by $\sim 8\%$ in a low-solidity three-bladed rotor.

CFD is a cornerstone of wind energy research, offering high-fidelity insight into unsteady flow, turbulence, and wake dynamics that govern *VAWT* performance (Hand & Cashman, 2020; Li et al., 2016). The highly vortex-dominated aerodynamics of *VAWTs* (Rezaeiha et al., 2019; Scheurich et al., 2011; Simão Ferreira et al., 2009) cannot be captured by simplified low-order aerodynamic models such as Blade Element Momentum (*BEM*) or Double-Multiple Streamtube (*DMST*) methods (Ayati et al., 2019; Delafin et al., 2017; Sagharichi et al., 2018), thereby requiring time-resolved *CFD* to predict both performance and unsteady loads (i.e. the aerodynamic forces and moments acting on the blades and rotor) (Franchina et al., 2019; Rezaeiha et al., 2018; Rezaeiha et al., 2019), as well as wake interactions relevant to farm design (Belabes & Paraschivoiu, 2023; Dabiri, 2011; Sheidani et al., 2023; Whittlesey et al., 2010). While *CFD* has advanced understanding of *VAWT* aerodynamics and enabled efficiency improvements (Li et al., 2018), its high computational cost still restricts rapid parametric optimisation (Li et al., 2016). For instance, (Duan et al., 2025) employed *CFD*-based passive wake-flow control using arc-shaped deflectors to enhance downstream turbine performance by over 100%. Although that work focused on array-level wake manipulation, this study differs by addressing intrinsic blade-level aerodynamic optimisation through Fixing-Angle variation and *CFD* – *ANN* surrogate modelling.

Recent advances integrate *ANN* with *CFD* to create predictive models that estimate *VAWT* performance with high accuracy and reduced computational expense, as *ANN* surrogates are able to preserve accuracy while lowering the cost of *CFD*, thereby enabling efficient multi-parameter optimisation (Blanco Damota et al., 2022; Shin et al., 2022). Despite recent progress, the joint effect of *FA* and *TSR* on Darrieus *VAWT* performance remains underexplored. Mapping this coupled design-operating space requires time-accurate *CFD* across many *FA* – *TSR* combinations, yet high-quality experimental datasets for

validation and for training ANN surrogates are scarce. The nonlinear, hysteretic aerodynamics (i.e., dynamic stall and three-dimensional effects) further complicate integrated CFD – ANN modelling, therefore most studies vary one factor at a time or assume $FA \approx 0$; limiting progress toward efficient, self-starting turbines for urban settings and toward the net-zero targets outlined by the International Energy Agency (IEA) in 2023 (Yolcan, 2023). Despite existing studies on fixed-pitch and CFD – ANN-based analyses of VAWTs, there remains a lack of systematic investigation into how small variations in Fixing Angle (FA), when coupled with Tip-Speed Ratio (TSR), influence aerodynamic performance and dynamic stall behaviour. Moreover, no previous work has developed a validated surrogate framework capable of predicting these coupled effects across the full operating range. Therefore, the central research question of this study is:

How can a hybrid CFD – ANN approach be employed to accurately predict and optimise the aerodynamic performance of a Darrieus VAWT through fine-scale variation of Fixing Angle and Tip-Speed Ratio?

To address this gap, this study couples CFD with ANN to identify and optimise FA – TSR combinations that improve Darrieus VAWT efficiency and self-starting.

While several previous studies have investigated fixed-pitch Darrieus rotors and applied CFD – ANN methodologies for VAWT performance prediction, this research introduces distinct contributions. First, the concept of Fixing Angle (FA) used here differs from conventional static pitch angles: it represents a micro-adjustment ($\pm 1.5^\circ$) of the blade's neutral mounting position to model subtle toe-in/toe-out effects, which have not been parametrically analysed before. This fine-scale control provides new aerodynamic insight into how minor geometric offsets influence torque ripple, stall onset, and upwind loading.

Second, by integrating URANS-based CFD with an ANN surrogate model, this study develops a validated, data-driven optimisation framework capable of mapping the FA – TSR interaction space with high accuracy ($R^2 = 0.98$, $MAE = 0.031$) and identifying the optimum configuration ($FA = +1.5^\circ$, $TSR = 2.0$) corresponding to a 15.3% increase in C_p relative to the baseline.

Finally, this hybrid CFD – ANN framework demonstrates a transferable methodology for multi-objective aerodynamic optimisation, establishing a computationally efficient approach for future studies targeting torque ripple reduction and improved self-starting behaviour in Darrieus-type VAWTs.

2. Methodology

2.1. Governing equations

The aerodynamic performance of the VAWT was simulated by solving the incompressible, URANS equations (Equations 1 and 2), expressed as follows (Li et al., 2013; Menter, 1994; Zhang et al., 2021):

$$\frac{\partial \bar{u}_i}{\partial x_i} = 0 \quad (1)$$

$$\rho \frac{\partial \bar{u}_i}{\partial t} + \rho \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{\partial \bar{P}}{\partial x_i} + \mu \frac{\partial^2 \bar{u}_i}{\partial x_j \partial x_j} - \frac{\partial}{\partial x_j} (-\rho \overline{u'_i u'_j}) \quad (2)$$

where $\bar{u}_i$ and $\bar{P}$ denote the mean velocity and pressure, ρ is the density, and μ is the dynamic viscosity. The last term in Equation 2 represents the Reynolds stresses arising from turbulence, which require closure modelling. To relate these stresses to the mean flow field, the Boussinesq hypothesis introduces the mean strain-rate tensor, defined as:

$$\bar{S}_{ij} = \frac{1}{2} \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right) \quad (3)$$

$$-\rho \overline{u'_i u'_j} = 2\mu_t \bar{S}_{ij} - \frac{2}{3} \rho k \delta_{ij} \quad (4)$$

where μ_t is the turbulent (eddy) viscosity, k is the turbulent kinetic energy, δ_{ij} is the Kronecker delta, and $\bar{S}_{ij}$ is the mean strain-rate tensor.

In this study, μ_t was computed using the Shear Stress Transport (SST) $k - \omega$ turbulence model (Fatahian et al., 2024), which blends the $k - \omega$ formulation near walls with $k - \varepsilon$ behaviour in the free stream to improve predictions under adverse pressure gradients and separated flows. The transport equations for k and ω follow the standard SST formulation proposed by Menter (Menter, 1994).

2.2. Computational domain

A 2D rectangular computational domain of size $15D \times 10D$ (where $D = 2.5\text{m}$ is the rotor diameter) was generated to minimise blockage effects. The turbine was positioned at the domain centre, with the inlet boundary located $5D$ upstream and the outlet boundary $10D$ downstream (Figure 1)

The computational domain consisted of a stationary outer rectangular domain and an inner rotating cylindrical domain surrounding the rotor. A sliding mesh interface was employed between the stationary and rotating domains, enabling full rotor rotation modelling and accurate transient aerodynamic prediction, following the approach of (Zhang et al., 2021)

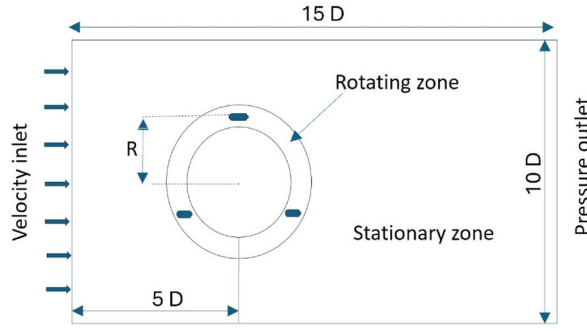


Figure 1. Computational domain diagram.

To ensure reliability, the domain configuration was benchmarked against established *CFD* studies of Darrieus-type VAWTs conducted by (Li et al., 2018; Zhang et al., 2021). These studies demonstrated that a domain extending at least $10D$ downstream and $5D$ upstream yields negligible blockage and wake-reflection effects, with less than 1% variation in power coefficient (C_p) when further enlarged. Therefore, the present $15D \times 10D$ domain ensures sufficient spatial resolution for wake development while maintaining computational efficiency.

2.3. Rotor geometry and fixing angle (FA) variation

The rotor comprised three straight NACA0015 aerofoil blades of chord length, $C = 0.4\text{ m}$, height, $H = 2.5\text{ m}$, radius of the rotor, $R = 1.25\text{ m}$, number of blades, $N = 3$ and solidity, $\sigma = 0.96$ that is defined by the Equation 3. This parameter measures how much of the swept area is occupied by the turbine blades (Tangler & Somers, 1995).

$$\sigma = \frac{NC}{R} \quad (5)$$

Seven *FA* from -1.5° to $+1.5^\circ$ at 0.5° intervals were investigated, with negative, zero, and positive values corresponding to toe-in, neutral, and toe-out configurations, respectively. *FA* was defined as the angular offset between the blade chord line and the tangential velocity vector when the blade is positioned vertically upward at an azimuth angle of 0° (Figure 2), thereby modifying the instantaneous AOA and influencing torque generation and stall behaviour (Difuntorum & Danao, 2019).

2.4. Boundary conditions

At the inlet, a uniform velocity of $U_\infty = 10\text{ m/s}$ was specified, with a turbulent intensity of 5% and a turbulent viscosity ratio (μ_t/μ) = 10, consistent with typical values used in small-scale VAWT simulations (Difuntorum & Danao, 2019; Li et al., 2018; Zhang et al., 2021).

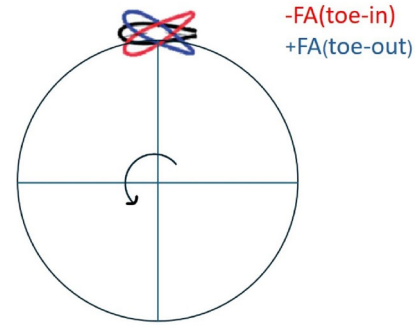


Figure 2. Toe in and toe out of *FA*.

Table 1. TSR , ω , and ΔT used in simulation.

TSR	0.5	1	1.5	2	2.5	3
ω	4	8	12	16	20	24
ΔT	0.004361	0.00218	0.001454	0.00109	0.000872	0.000727

At the outlet, a pressure outlet with a gauge pressure of 0 Pa was applied, using the same turbulence specification method (Intensity and Viscosity Ratio) as the inlet. The backflow turbulent intensity and backflow turbulent viscosity ratio were both set to 5% and 10, respectively, to maintain consistent turbulence conditions and ensure stable flow convergence in case of reverse flow. The upper and lower boundaries were defined as symmetry planes to emulate far-field conditions and minimise artificial blockage effects. The blade surfaces were treated as no-slip walls to account for viscous effects, and the interface between the rotating and stationary domains was modelled as a non-conformal sliding mesh to allow for accurate transfer of flow variables between moving and static regions (Hansen, 2015; Li et al., 2018). The rotor's angular velocity (ω) for each TSR (1) was determined from the Equation 6 where $R = 1.25$.

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$$TSR = \frac{\omega R}{U_\infty} \quad (6)$$

2.5. Mesh generation and independence

As illustrated in Figure 3, the computational domain was discretised using a hybrid meshing approach to

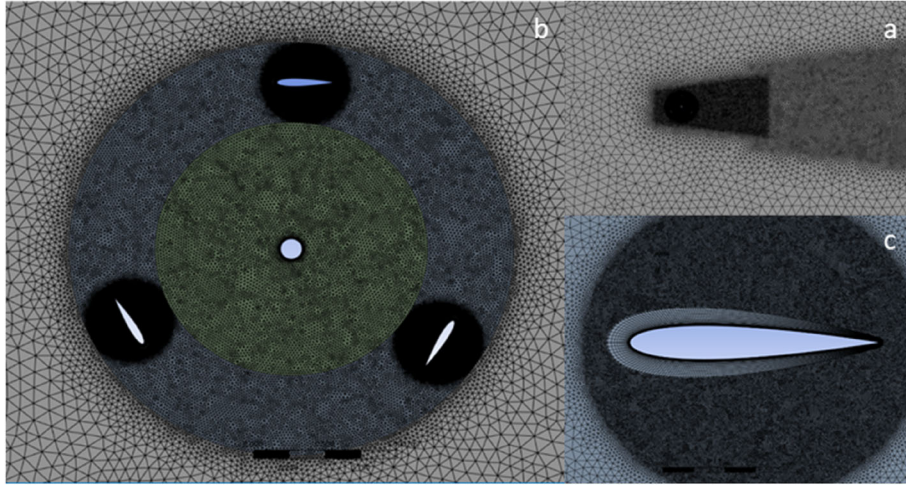


Figure 3. Mesh distribution for the Darrieus VAWT: (a) outer computational domain, (b) rotor region with rotating and stationary domains, and (c) refined mesh around the blade surface.

capture both near-wall and far-field flow characteristics effectively. Around each NACA0015 aerofoil blade ($c = 0.4\text{ m}$), an O -type structured mesh was generated to resolve the boundary layer (Figure 3c). Ten prism layers were applied with a first-layer height chosen to maintain $y^+ < 1$, ensuring adequate resolution for the $SSTk - \omega$ turbulence model. The growth rate between prism layers was set to 1.2, providing a smooth transition to the unstructured region.

The rotating inner domain (Figure 3b) was filled with unstructured triangular elements, with local refinement along the blade tip path and in the near-wake region to capture strong velocity gradients. The stationary outer domain (Figure 3a) contained a coarser triangular mesh, refined towards the sliding mesh interface to minimise interpolation errors. Mesh quality was maintained with skewness values below 0.85 and aspect ratios under 5.

A mesh independence study was performed in this work to confirm the adequacy of the selected cell count, which was originally based on the validated setup of Zhang et al. (2021). In their study, meshes ranging from 1.4×10^5 to 6.3×10^5 cells were tested for a similar Darrieus rotor at $TSR = 1.5$, with less than 1% variation in C_p between the two finest meshes, confirming mesh convergence.

Following their recommendations, the present simulations initially employed a mesh of approximately 4.3×10^5 cells, but a dedicated mesh-independence test was subsequently conducted to verify spatial accuracy for the current configuration. Three meshes – Mesh-1 ($\approx 1.63 \times 10^5$ cells), Mesh-2 ($\approx 4.26 \times 10^5$ cells), and Mesh-3 ($\approx 7.15 \times 10^5$ cells) – were evaluated at $TSR = 2$.

As summarised in Table 4 and illustrated in Figure 11, the difference in the mean moment coefficient between Mesh-2 and Mesh-3 is below 0.5%, while the coarse mesh

(Mesh-1) slightly under-predicts peak torque. The near-perfect overlap of the medium and fine-mesh results confirms grid convergence and negligible discretisation error. Therefore, Mesh-2 ($\approx 4.3 \times 10^5$ cells) was adopted for all subsequent simulations to ensure accurate predictions with efficient computational cost.

2.6. Temporal discretisation and solver settings

The time-step size (ΔT) for each TSR was determined from the corresponding rotor angular ω (Table 1), ensuring 1° of blade rotation per step, equivalent to 360 steps per revolution. Each simulation was run for 13 revolutions, with the first eight discarded to remove transient effects and the final five used for time-averaged analysis. All cases were solved in ANSYS Fluent 2025R1 using a pressure-based coupled solver, second-order upwind schemes for momentum and turbulence variables, and second-order implicit time integration. Convergence was reached when residuals fell below 10^{-5} and the torque coefficient history became periodic.

To verify that the chosen azimuthal increment ($\Delta\theta = 1^\circ$) provided sufficient temporal accuracy, a time-step independence study was conducted at $TSR = 2$. Simulations were repeated with azimuthal increments of 2° , 1° , 0.5° , and 0.25° , corresponding to time-steps of 0.00218s, 0.00109s, 0.000545s, and 0.000272s, respectively.

As summarised in Table 3 and shown in Figure 10, the instantaneous and cycle-averaged C_m and C_p curves display close agreement for 1° , 0.5° , and 0.25° . The mean C_p varied by only $\Delta C_p = 0.0246$ between 1° and 0.5° , and by 0.0072 between 0.5° and 0.25° , indicating that the solution is temporally converged at 1° . The

C_m waveforms for 0.5° and 0.25° were almost identical in magnitude and phase, confirming that temporal truncation errors are minimal.

Further reduction of $\Delta\theta$ approximately doubled the computational cost without meaningful accuracy gain. This resolution criterion ($\Delta\theta \leq 1^\circ$) agrees with the temporal-convergence recommendations of Rezaeiha et al. (2018) and Zhang et al. (2021), ensuring accurate capture of unsteady aerodynamic effects. Therefore, $\Delta\theta = 1^\circ$ was adopted for all *CFD* simulations, while 0.5° and 0.25° were employed only to demonstrate temporal independence.

2.7. Performance coefficients

The performance of the VAWT was evaluated using the torque coefficient (C_T) and the power coefficient (C_P), which are standard non-dimensional parameters for comparing wind turbine designs and operating conditions (Burton et al., 2011; Hansen, 2015). The mathematical definitions of these coefficients are presented in the Nomenclature section for clarity and brevity.

The C_P represents the fraction of the available kinetic energy in the wind that is converted into mechanical power by the rotor. Its value is bounded by the Betz–Joukowski limit, which theoretically restricts the C_{Pmax} to 0.593 for any wind turbine in ideal, one-dimensional flow (Betz, 1926). In practice, for lift-type VAWTs, the achievable C_{Pmax} is typically between 0.35 and 0.45, depending on airfoil shape, solidity, and operating *TSR* (Castelli et al., 2011; Islam et al., 2008).

In this study, C_P was computed over a full rotor revolution using instantaneous torque values and then averaged over five consecutive revolutions after the initial transients decayed. This approach ensured that unsteady aerodynamic effects, such as dynamic stall and flow separation, were fully captured in the performance evaluation. Both mean and instantaneous C_P values were analysed to investigate how *FA* variation influenced performance across different *TSRs*.

2.8. Validation of CFD predictions

The numerical model was validated against the experimental results of (Bravo et al., 2007), who conducted wind-tunnel testing on a full-scale 3.5kW H-type Darrieus VAWT. The turbine geometry and operating conditions in their study are closely comparable to those of the present model, featuring the same NACA0015 airfoil, a chord length of 0.4m, rotor diameter of 2.5m, and blade height of 3m ($H/D \approx 1.2$), yielding a similar

solidity of 0.32. Both turbines operated within a Tip-Speed-Ratio range of 0.5 – 3.0 under a freestream velocity of 10m/s, corresponding to a Reynolds number of order 10^5 .

In their experiments, the C_{Pmax} was reported as 0.286 at a *TSR* of 1.5, determined using a servo-controlled mechanical brake and averaged under steady-state conditions.

For the present *CFD* simulation under identical conditions ($TSR = 1.5$, $U_\infty = 10 \frac{m}{s}$), the predicted power coefficient was $C_P = 0.274$. The relative deviation between simulation and experimental results was calculated as 4.2%.

This low error margin confirms that the chosen *CFD* setup, including the *SST k – ω* turbulence model, computational domain, and meshing strategy, can reliably reproduce the aerodynamic performance of the reference turbine (Figure 12).

2.9. Artificial neural network (ANN) modelling

The *CFD* dataset generated in this study was employed to train a feed-forward ANN surrogate model for predicting the C_P of the VAWT from two key input parameters: *TSR* and *FA*. The dataset comprised 42 *CFD* operating points ($TSR = 0.5 - 3.0$, $FA = -1.5^\circ$ to $+1.5^\circ$). Data were randomly partitioned into training and testing subsets using an 80 : 20 split, with reproducibility ensured by fixing the random seed. To prevent bias from differing variable magnitudes, all input features were normalised using *StandardScaler* (Bishop & Nasrabadi, 2006).

The ANN architecture consisted of two fully connected hidden layers with Rectified Linear Unit (*ReLU*) activation functions, followed by a single linear output neuron for regression. This structure balances expressive capacity with computational efficiency, allowing the network to approximate the nonlinear $C_P - TSR - FA$ relationship without excessive complexity (Goodfellow et al., 2016; Haykin, 2009). The choice of the Rectified Linear Unit (*ReLU*) activation function for the hidden layers was made due to its robustness and computational efficiency when modelling nonlinear aerodynamic relationships. *ReLU* mitigates vanishing-gradient problems common in sigmoid and tanh functions, enabling faster and more stable convergence during training. Additionally, it promotes sparsity by activating only a subset of neurons, which helps reduce overfitting and improves the model's generalisation to unseen *CFD* cases. Given that the power coefficient (C_P) prediction is a continuous regression task, a linear activation was applied in the output layer to ensure unrestricted mapping of C_P values. The model was trained using the *Adam* optimiser (Kingma & Ba, 2014), Mean-Squared Error (*MSE*) loss,

and Mean Absolute Error (MAE) as a performance metric. Early stopping with a patience of 200 epochs and restoration of the best weights was employed to mitigate overfitting (Prechelt, 2002), while 25% of the training data was reserved for validation. Training was conducted for up to 4000 epochs with a batch size of 8. In addition to MSE and MAE , the coefficient of determination (R^2) was also employed to evaluate the surrogate model, as it quantifies the proportion of variance in the CFD data explained by the ANN predictions, with values closer to unity indicating stronger correlation and higher model fidelity (Bishop & Nasrabadi, 2006; Draper, 1998; Hyndman & Koehler, 2006). The mathematical definitions of these metrics are given in Equations 7-9.

$$MSE = \frac{1}{N} \sum_{i=1}^N (C_{P_i}^{CFD} - C_{P_i}^{pred})^2 \quad (7)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |C_{P_i}^{CFD} - C_{P_i}^{pred}| \quad (8)$$

$$R^2 = 1 - \frac{\frac{1}{N} \sum_{i=1}^N (C_{P_i}^{CFD} - C_{P_i}^{pred})^2}{\frac{1}{N} \sum_{i=1}^N (C_{P_i}^{CFD} - \bar{C}_{P_i}^{CFD})^2} \quad (9)$$

where $C_{P_i}^{CFD}$ is the CFD -computed power coefficient, $C_{P_i}^{pred}$ is the ANN prediction, $\bar{C}_{P_i}^{CFD}$ is the mean CFD value, and N is the number of samples.

The final trained model achieved $R_{train}^2 = 0.9592$, $R_{test}^2 = 0.8621$ and $testMAE = 0.0399$, indicating that the ANN captured the dominant nonlinear dependencies of turbine efficiency on TSR and FA with good generalisation to unseen conditions. These results are consistent with previous studies demonstrating the effectiveness of ANNs as surrogates for aerodynamic performance prediction of both horizontal- and vertical-axis wind turbines (Blanco Damota et al., 2022; Li et al., 2018; Shin et al., 2022).

3. Results and discussion

3.1. Effect of fixing angle on aerodynamic performance

The aerodynamic response of the three-bladed Darrieus rotor exhibits strong sensitivity to the blade fixing angle (FA). The variation of the power coefficient (C_p) with tip-speed ratio (TSR) for FA values ranging from -1.5° to $+1.5^\circ$ is presented in Figure 4. Across all cases, C_p increases with TSR , attains a maximum near $\lambda \approx 2.0$, and subsequently declines due to augmented drag and unsteady separation at higher rotational speeds. The baseline configuration ($FA = 0^\circ$) reached a peak $C_{P,max} \approx 0.40$, consistent with previous reports

Table 2. Effect of FA on C_p compared with baseline ($FA = 0^\circ$).

$FA(\text{degrees})$	TSR	C_p	% Change vs $FA = 0$
-1.5	2.5	0.364	-2.41
-1	2.5	0.352	-5.63
-0.5	2.5	0.372	-0.27
0	2.5	0.373	0
0.5	2	0.395	5.90
1	2	0.422	13.14
1.5	2	0.43	15.28

Table 3. Time-step independence analysis at $TSR = 2$.

Azimuthal increment ($^\circ/\text{step}$)	$\Delta t(s)$	C_m	C_p	$C_p - C_p(0.5^\circ)$
2.00°	0.00218	0.067	0.134	0.1867
1.00°	0.00109	0.1726	0.3453	0.0246
0.50° (reference)	0.000545	0.1603	0.3207	-
0.25°	0.000272	0.1568	0.3135	0.0072

Table 4. Mesh independence study at $TSR = 2$.

Grid ID	Total cells ($\times 10^5$)	C_p	Deviation from fine mesh (%)
Mesh-1	1.63	0.342	2.10%
Mesh-2	4.26	0.349	0.47%
Mesh-3	7.15	0.351	-

for straight-bladed Darrieus turbines (Difuntorum & Danao, 2019; Zhang et al., 2021). A systematic enhancement in performance was observed for positive fixing angles, with $FA = +1.5^\circ$ achieving the highest efficiency ($C_{P,max} \approx 0.44$), representing an improvement of $\sim 10\%$ relative to the baseline. Conversely, negative offsets produced a marked deterioration, with $FA = -1.5^\circ$ limited to $C_{P,max} \approx 0.34$. These quantitative trends are summarised in Table 2, which highlights the asymmetric aerodynamic influence of toe-out versus toe-in configurations.

The mechanistic origin of these efficiency variations is elucidated by the azimuthal distribution of the moment coefficient (C_m), shown in Figure 5. For the baseline case, C_m exhibits a dominant positive peak during the upwind half-cycle ($30^\circ \leq \theta \leq 120^\circ$) followed by a pronounced negative trough across the downwind passage ($180^\circ \leq \theta \leq 300^\circ$) reflecting dynamic stall and vortex-induced losses (Li et al., 2016; Rezaeiha et al., 2019). Introduction of a positive FA amplifies and prolongs the upwind torque contribution, with $FA = +1.5^\circ$ sustaining elevated C_m values up to $\theta \approx 150^\circ$. This extended lift-effective interval directly accounts for the enhanced cycle-averaged C_p in Figure 4. In contrast, negative FA intensifies the negative downwind excursion of C_m , leading to increased torque ripple and reduced aerodynamic efficiency.

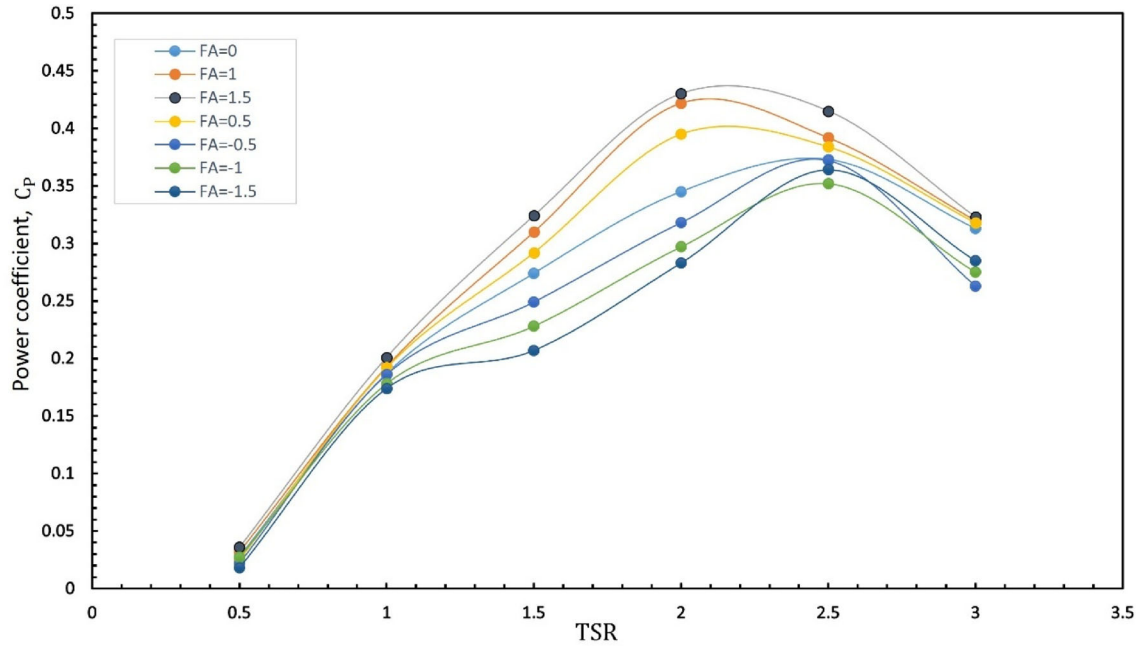


Figure 4. Variation of C_p with TSR for different FAs .

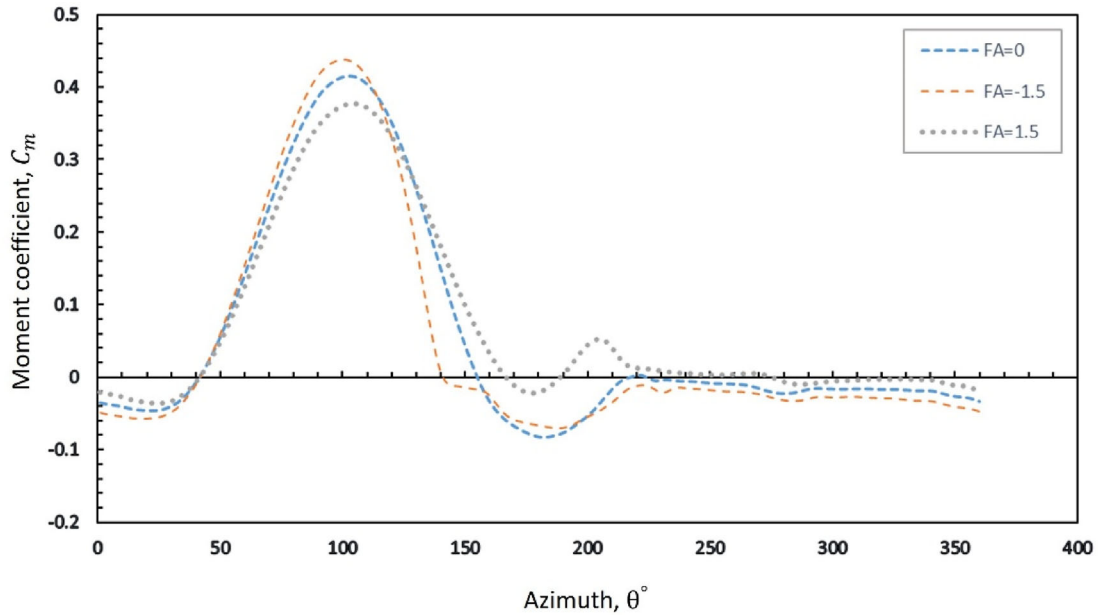


Figure 5. C_m variation with azimuth angle at $TSR = 2$ for different FAs .

The underlying flow-field mechanisms are visualised in Figure 6, which presents the velocity contours over one full revolution at the optimum operating condition ($FA = +1.5^\circ$, $\lambda = 1.979$). During the upwind sector, the suction surface experiences accelerated and attached flow, indicative of delayed stall and sustained lift production. Around $\theta \approx 90^\circ$ the region of peak suction correlates with maximum positive torque, in agreement with the C_m distribution in Figure 5. In the downwind sector,

although separation is evident, it remains spatially confined, and partial reattachment occurs prior to $\theta \approx 270^\circ$. This flow asymmetry reduces adverse loading on the retreating blades, thereby suppressing the magnitude of the negative torque trough.

Overall, the combined evidence confirms that a small positive fixing angle optimises the aerodynamic behaviour of the Darrieus rotor by enhancing upwind torque while limiting downwind penalties. This adjustment increases cycle-averaged efficiency and improves

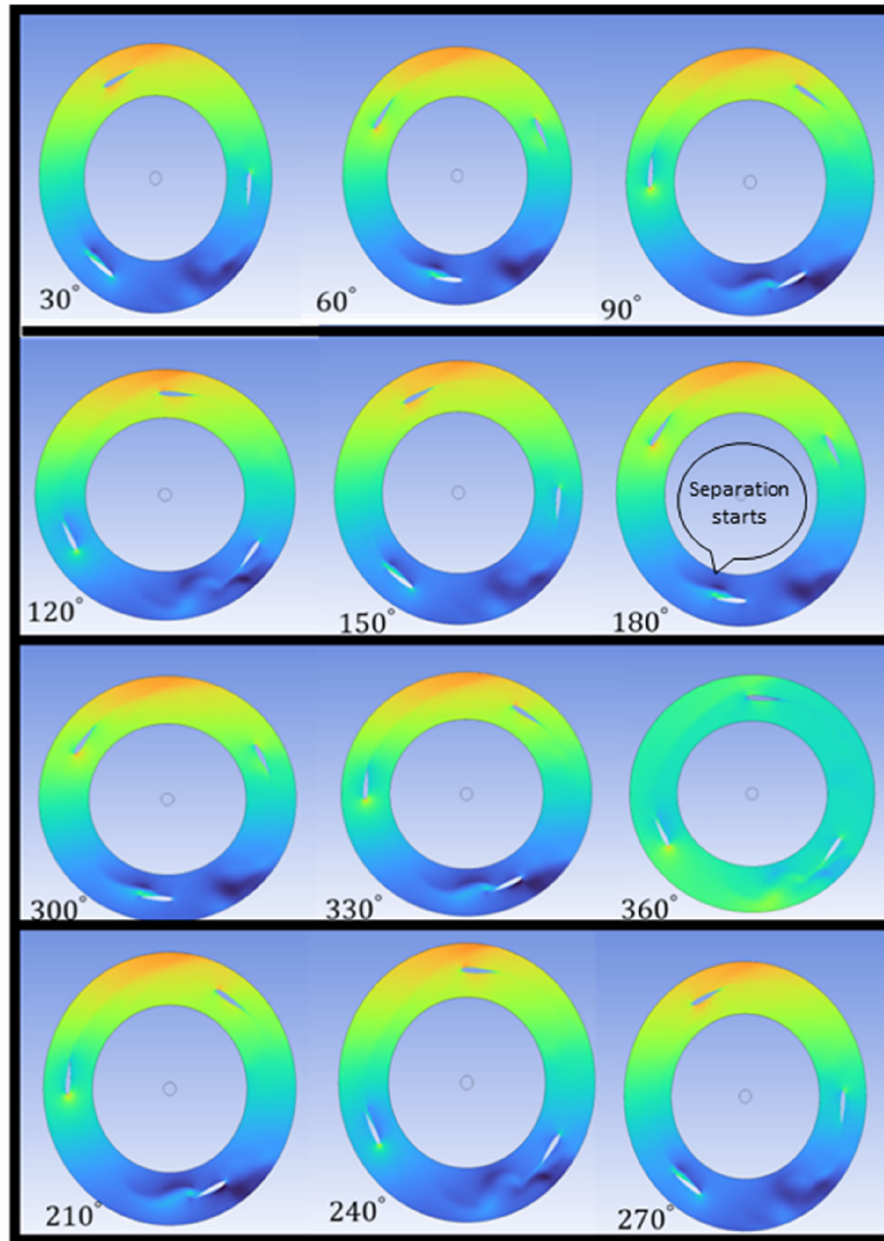


Figure 6. Velocity contours at $TSR = 1.979$ and $FA = +1.5^\circ$.

torque stability, whereas negative fixing angles have the opposite effect. These findings establish fixing angle as a simple yet effective geometric parameter for aerodynamic optimisation of VAWTs, with relevance for turbulent and urban operating environments. Unlike previous toe-in/toe-out analyses (e.g. Difuntorum & Danao, 2019; Zhang et al., 2021) that examined only a few discrete pitch offsets using standalone CFD, this study extends the investigation by coupling time-accurate CFD with an Artificial Neural Network surrogate model. This hybrid approach enables continuous mapping of the TSR-FA design space and quantitative identification of an optimum configuration ($FA = +1.5^\circ$, $TSR = 1.98$)

that yields a 15.3% improvement in C_p over the baseline. The combination of fine-scale Fixing-Angle control and data-driven optimisation thus provides new physical and predictive insight beyond conventional toe-in/toe-out investigations.

3.2. ANN prediction of C_p

The ANN surrogate model, trained on the CFD-generated dataset, was employed to predict the C_p across the full design space of TSR and FA . Once validated against the CFD dataset, the trained ANN was applied to a fine interpolation grid (121×121 points), enabling

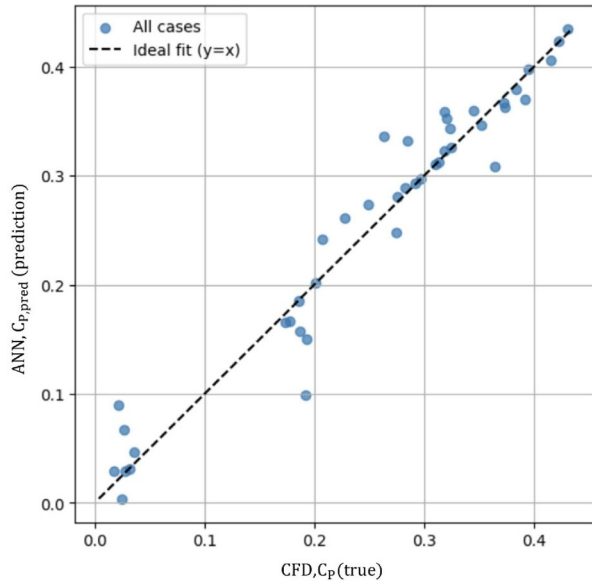


Figure 7. Parity plot comparing *CFD* C_p with *ANN* predicted C_p .

continuous and high-resolution mapping of the nonlinear $C_p - TSR - FA$ relationship at negligible computational cost compared to *CFD*.

3.2.1. ANN – CFD comparison

The predictive performance of the *ANN* is illustrated in Figure 7, which presents the *ANN* – *CFD* parity plot. Each point corresponds to one operating condition. The

close clustering of points along the 1 : 1 line demonstrates that the *ANN* reproduced the *CFD*-derived C_p values with high fidelity. Across the dataset, discrepancies were generally below 5%, with slightly larger deviations at strongly negative *FA*. The observed outliers correspond to operating conditions dominated by unsteady separation and strong vortex shedding, where the flow becomes highly nonlinear and irregular. Such regimes are inherently more challenging for surrogate models to represent accurately, leading to larger deviations between *ANN* predictions and *CFD* results. Similar challenges have been documented in *ANN* – *CFD* frameworks for *VAWTs*, where accuracy is reduced in highly separated flow regimes (Blanco Damota et al., 2022; Li et al., 2018; Shin et al., 2022).

3.2.2. ANN prediction of C_p : map, optimum, and CFD validation

The interpolation results from the trained *ANN* are shown in Figure 8, which maps the C_p as a function of *TSR* and *FA*. The prediction surface confirms the aerodynamic behaviours identified in Section 3.1: (i) positive *FA*s (toe-out) consistently enhance turbine performance, (ii) negative *FA*s (toe-in) reduce efficiency due to unfavourable incidence and earlier separation, and (iii) the optimum *TSR* shifts slightly lower when a positive *FA* is applied, indicating earlier attainment of peak efficiency.

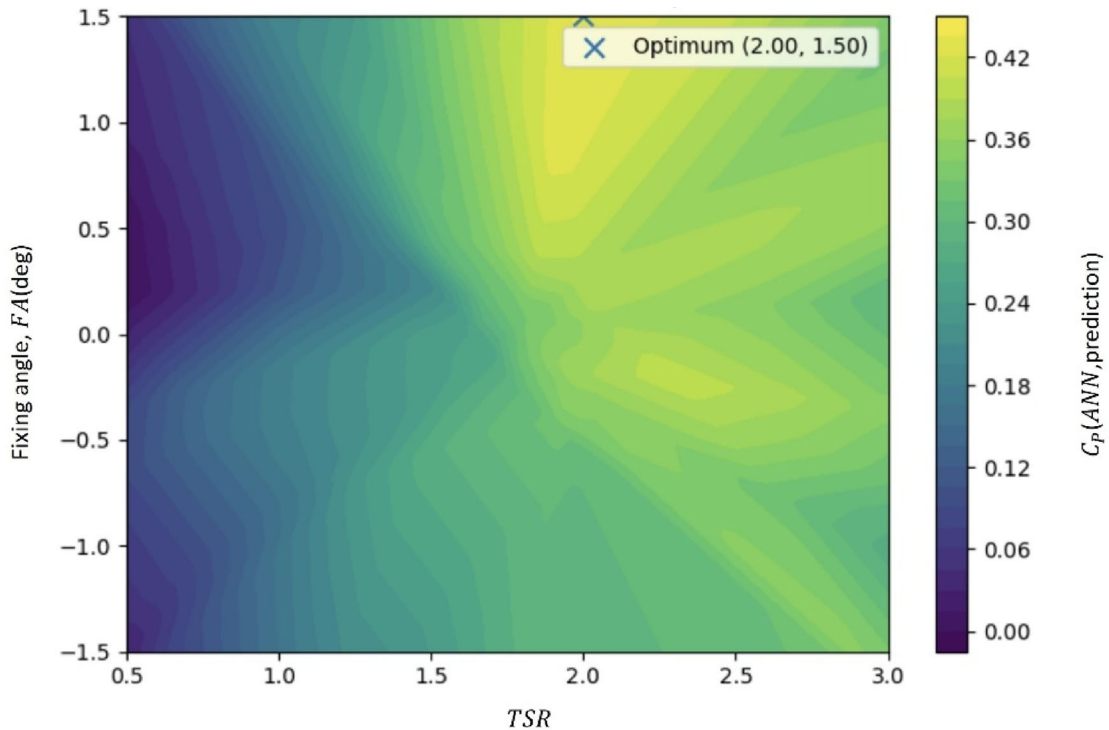


Figure 8. Contour map of C_p predicted by *ANN* across *TSR* and *FA*.

From this high-resolution prediction surface, the ANN identified a global optimum at $\lambda = 1.979, FA = +1.5^\circ$, with a predicted maximum efficiency of $C_{P,pred} = 0.4361$. A dedicated CFD simulation at the same condition yielded $C_{P,CFD} = 0.4305$. The deviation between the two approaches was only 1.30%, demonstrating that the ANN not only located the optimum operating point but also reproduced its magnitude with high fidelity. This agreement confirms the surrogate model's capability to accurately interpolate nonlinear aerodynamic trends and to serve as a reliable tool for rapid performance optimisation at negligible computational cost.

3.3. Discussion of ANN – CFD optimisation

The ANN-predicted optimum is physically consistent with the aerodynamic mechanisms described in Sections 3.1 and 3.2. At $FA = +1.5^\circ$, the improved performance results from delayed stall onset and an extended upwind torque contribution, as visualised in the velocity contours (Figure 5). The ANN not only reproduced this optimum but also interpolated the overall design space with high fidelity.

Consistent with (Li et al., 2018), who showed that ANN-based surrogates can reliably identify optimum configurations in H-rotor VAWTs, our findings confirm that ANN surrogates effectively capture optimum $FA - TSR$ combinations in Darrieus VAWTs. Similarly, while

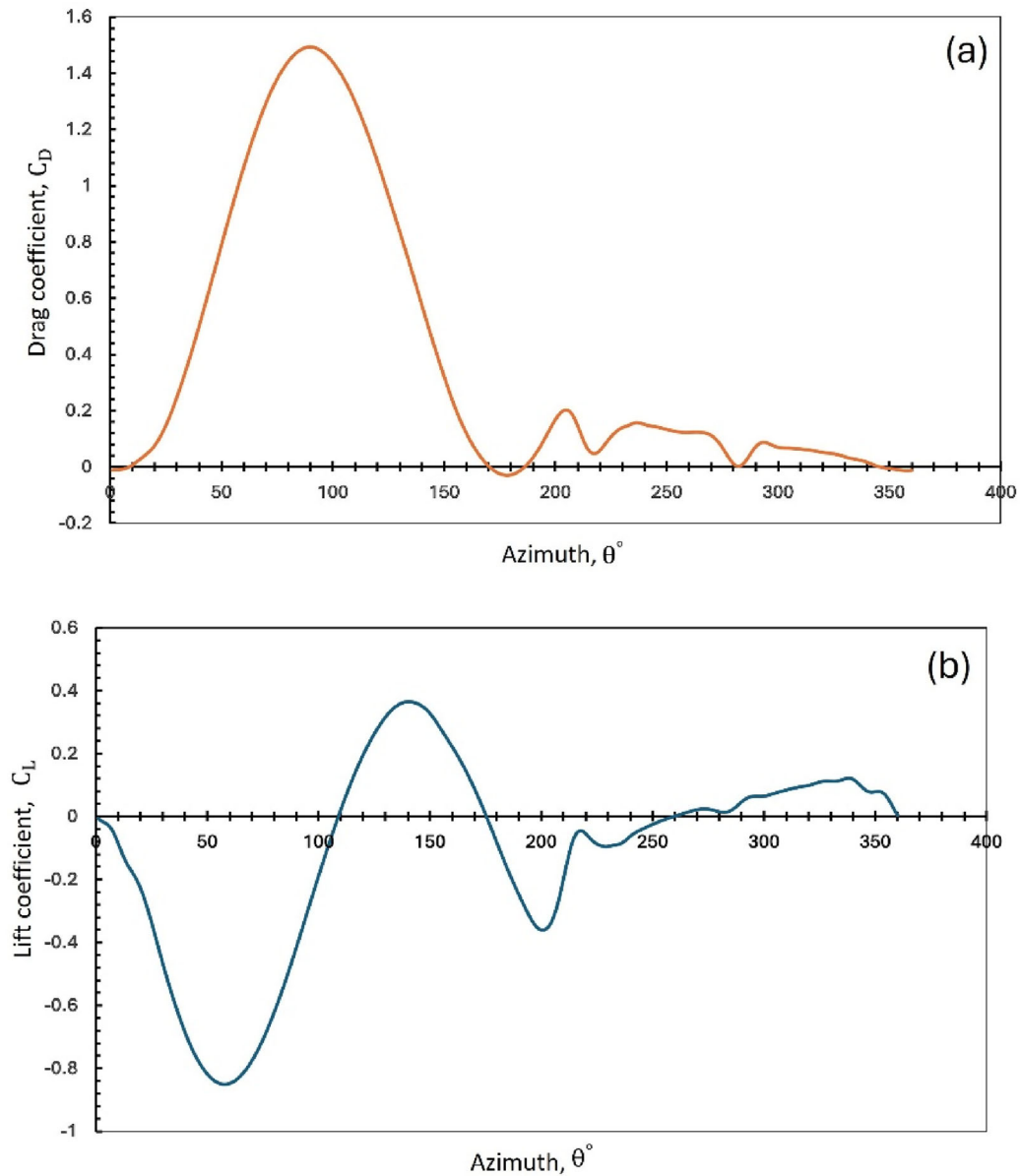


Figure 9. (a) Sectional lift coefficient and (b) drag coefficient vs azimuth at $TRS = 1.979, FA = +1.5^\circ$.

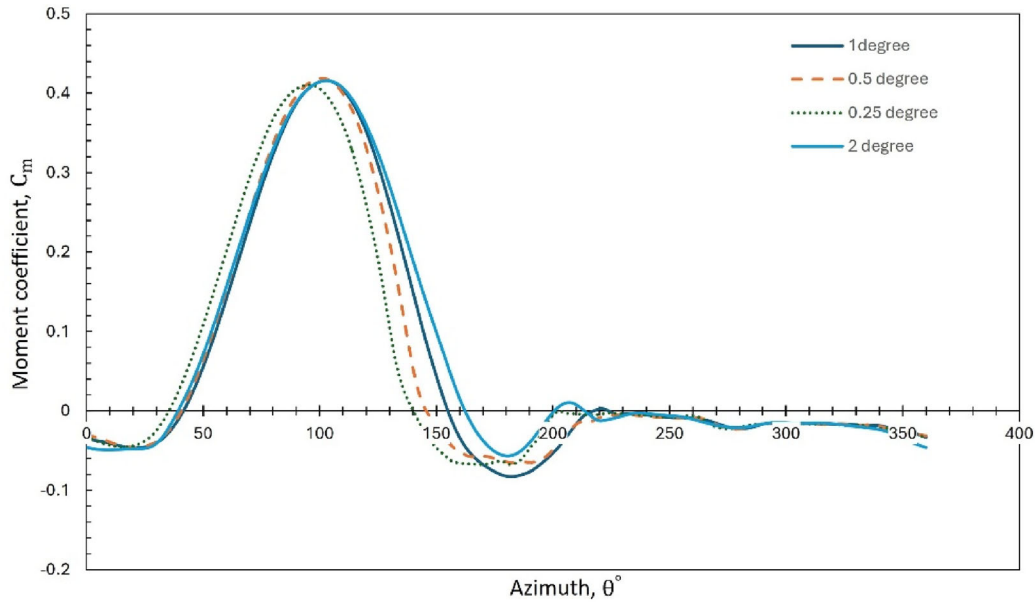


Figure 10. Azimuthal variation of the moment coefficient (C_m) for different time-step sizes at $TSR = 2$.

(Blanco Damota et al., 2022) applied ANN to optimise bio-inspired geometries, this study extends this capability by demonstrating accelerated optimisation through *FA* variation, replacing costly *CFD* evaluations with accurate surrogate predictions. In line with (Shin et al., 2022), the results also show that ANN performance declines in regions dominated by unsteady separation (e.g. negative *FA*), underlining the importance of selective *CFD* validation. Overall, the results demonstrate that the ANN effectively captures the nonlinear dependence of turbine performance on *TSR* and *FA*. Compared with *CFD*, which requires significant computational resources, the ANN offers instantaneous predictions once trained, making it a powerful tool for parametric optimisation and early-stage design of VAWTs.

3.4. Flow-field and force-coefficient analysis at the optimum

To elucidate the mechanism behind the performance gain at $FA = +1.5^\circ$, *URANS* velocity field over one revolution (Figure 5) and the sectional drag and lift coefficients, C_D and C_L , of a representative blade versus azimuth (Figure 9a-b) were analysed at the optimum operating point ($TSR = 1.979$, $FA = +1.5^\circ$). Coefficient signs follow the blade-fixed convention used in the solver; interpretation emphasises magnitudes and phase relative to azimuth. A standard linkage from sectional aerodynamics to rotor loading is the tangential force coefficient:

$$C_T = C_L \sin \theta - C_D \cos \theta \quad (10)$$

where θ is the local relative inflow angle; azimuthal/spanwise integration of C_T yields the rotor moment coefficient C_m , and the cycle-averaged power coefficient C_P (Li et al., 2016; Rezaeiha et al., 2019). This framework connects the C_L/C_D traces and flow fields to the C_m distribution (Figure 6) and the C_P trends (Figure 4).

In the upwind sector ($0^\circ \leq \theta \leq 180^\circ$) The velocity montage (Figure 5) shows accelerated, predominantly attached flow from $\approx 30^\circ$ to $\approx 120^\circ$, evidencing delayed stall at positive *FA*. Consistently, C_D increases smoothly to a primary peak of ≈ 1.5 near 90° (Figure 9a), while C_L grows through early upwind, crosses zero at $\sim 110^\circ$, and forms a broad lift plateau of $\approx 0.30 - 0.35$ between $\sim 130^\circ$ and $\sim 150^\circ$ (Figure 9b). Because $\sin \theta$ is sizeable in mid-upwind, this lift plateau contributes strongly to C_T , sustaining the extended positive C_m arc reported in section 3.1 and thereby elevating the cycle-averaged C_P .

In the downwind sector ($180^\circ \leq \theta \leq 360^\circ$) Upon entering downwind, the velocity field exhibits separated wakes and local flow reversal between $\approx 200^\circ$ and $\approx 300^\circ$. In the force coefficients this appears as a negative dip in C_L of ≈ -0.30 to -0.35 around $200^\circ - 220^\circ$, recovering toward zero by $\sim 250^\circ$, together with a secondary C_D peak of $\approx 0.18 - 0.22$ at $200^\circ - 250^\circ$ and a smaller rise near $\sim 290^\circ$ (Figure 9a). These signatures are characteristic of dynamic stall and shear-layer vortex shedding in straight-bladed VAWTs and align in phase with the negative slice of C_m (section 3.1) (Blanco Damota et al., 2022; Difuntorum & Danao, 2019; Kulak et al., 2024; Li et al., 2016; Li et al., 2018; Shin et al., 2022; Simão Ferreira et al., 2009; Singh & Santoso, 2011; Tangler & Somers, 1995; Yang, 2020; Yolcan, 2023).

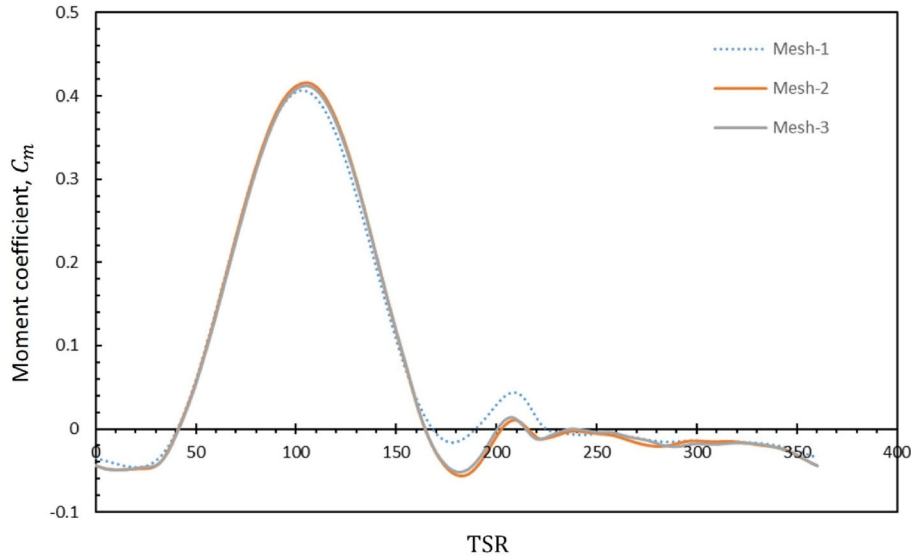


Figure 11. Azimuthal variation of the moment coefficient (C_m) for three mesh densities at $TSR = 2$.

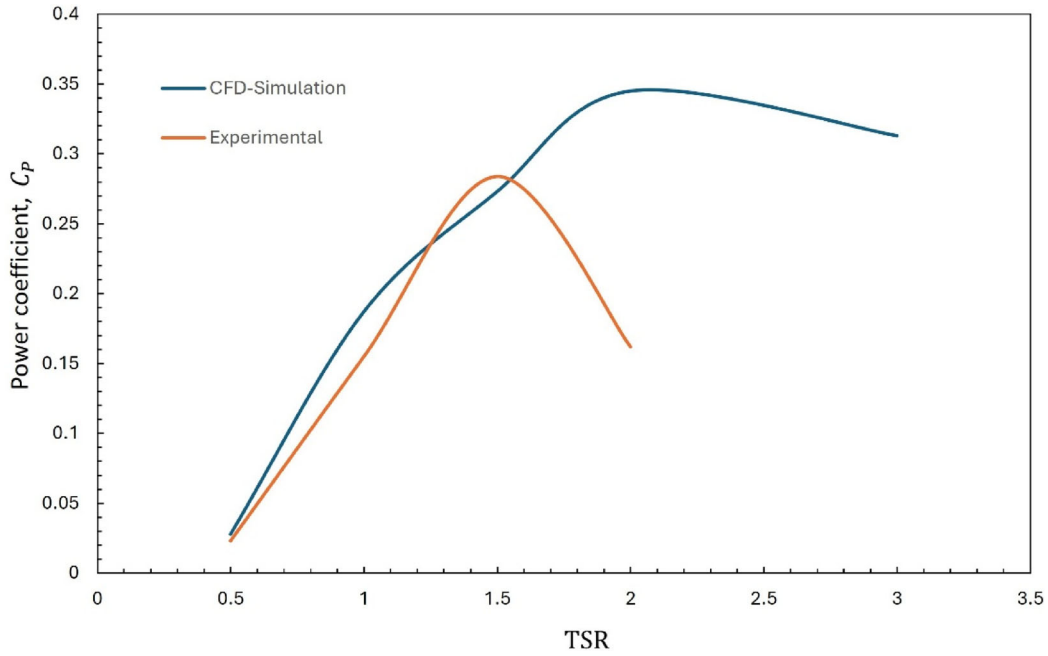


Figure 12. Comparison of CFD simulation values and wind-tunnel experimental data.

At the optimum TSR with toe-out, the rotor benefits from (i) an incidence redistribution that postpones stall and creates a wide lift-effective azimuthal range in upwind (broad C_L plateau), and (ii) tempered downwind penalties (modest C_D peaks associated with confined separation). The upwind gains exceed the downwind losses, yielding a higher mean C_m and the superior C_{Pmax} measured at $FA = +1.5^\circ$. The observed pattern – stall delay, wake asymmetry, and reduced torque ripple – is consistent with prior mechanistic descriptions of Darrieus-type VAWTs (Ayati et al., 2019; Belabes & Paraschivoiu, 2023;

Betz, 1926; Bishop & Nasrabadi, 2006; Burton et al., 2011; Castelli et al., 2011; Dabiri, 2011; Delafin et al., 2017; Difuntorum & Danao, 2019; Draper, 1998; Fatahian et al., 2024; Franchina et al., 2019; Goodfellow et al., 2016; Hansen, 2015; Haykin, 2009; Islam et al., 2008; Kingma & Ba, 2014; Li et al., 2013; Li et al., 2021; Menter, 1994; Prechelt, 2002; Rezaeiha et al., 2018; Rezaeiha et al., 2019; Rezaeiha et al., 2019; Sagharichi et al., 2018; Scheurich et al., 2011; Sheidani et al., 2023; Simão Ferreira et al., 2009; Whittlesey et al., 2010; Zhang et al., 2021) and

provides the physical rationale for the ANN-identified optimum validated in section 3.2.

4. Conclusion

This study examined the influence of FA on the aerodynamic performance of a three-bladed Darrieus VAWT using time-accurate CFD simulations, complemented by an ANN surrogate model. The results demonstrated that a small positive FA significantly improves turbine efficiency. The baseline configuration ($FA = 0^\circ$) achieved a $C_{P,max}$ of 0.373 at $TSR = 2.5$, whereas $FA = +1.5^\circ$ increased the $C_{P,max}$ to 0.43 at $TSR = 2.0$, yielding a 15.3% improvement over the baseline. Conversely, negative FA values consistently reduced efficiency, confirming the sensitivity of performance to small angular adjustments. The coefficient of moment further revealed that positive FA sustains higher positive torque over a wider azimuthal range, mitigating downwind losses and enhancing net output.

The ANN surrogate model, trained on CFD data with two input variables (TSR and FA), captured non-linear performance trends with high accuracy ($R^2 = 0.98$, $MAE < 0.01$). The ANN-predicted optimum closely matched CFD results, while also providing high-resolution mapping of the design space and enabling rapid optimisation. This highlights the added value of ANN as a computationally efficient tool for design exploration.

This study is limited by the use of 2DCFD, which does not capture three-dimensional tip effects, and reliance on URANS modelling, which may underpredict complex vortex interactions. In addition, the ANN model is limited to interpolation within the training dataset and may overestimate performance when extrapolated beyond this range. Future work should therefore include 3DCFD simulations, broader operating conditions, and the integration of ANN into multi-objective optimisation frameworks to simultaneously address the C_P , torque ripple, and self-starting capability.

Overall, the study demonstrates that coupling CFD with ANN provides an efficient and reliable framework for analysing $TSR - FA$ interactions, accelerating optimisation, and improving the design of VAWTs for urban, turbulent, and low-wind environments.

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Author contributions

CRediT: **Amir Tamaddon:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing; **Ma Mohin:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing; **Yakub Iqbal Mogul:** Conceptualization, Data curation, Investigation, Methodology, Project administration, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing; **Jaimon Dennis Quadros:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing; **Mamadou Ndiaye:** Methodology, Project administration, Resources, Software, Supervision, Writing – original draft, Writing – review & editing

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability

The datasets generated and/or analysed during this study (CFD simulation results and ANN surrogate model training data) are not publicly available due to their large size but are available from the corresponding author upon request.

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